

Aalekh Bukhariya

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WebPage | GitHub | LinkedIn

Research Interests

AI safety, pluralistic alignment, LLM response diversity, hallucination interpretability.

Education

George Washington University. Washington, D.C., USA 2025–2027

M.S. in Computer Science GPA: 4.0/4.0

Relevant Coursework: Design and Analysis of Algorithms; Computer System Architecture; Advanced Software Paradigms; Machine Learning; Neural Networks and Deep Learning; Robotics vision and perception

Indian Institute of Information Technology, Surat. Gujarat, India 2021–2025

B.Tech. in Computer Science and Engineering GPA: 8.48/10

Experience

Research Intern, IIT Indore, MODSS Lab Jan 2025–Jul 2025

Supervisor: Dr. Kapil Ahuja

Applied transformers, LSTMs, AdaHessian, and ensemble methods to DNA gap prediction, framing missing genomic regions as a masked sequence prediction problem. Studied how large language models can be adapted for genetic data generation via masked sequence modelling. Contributed to a manuscript on genomic sequence imputation currently in preparation.

Research

Mitigating the Hivemind: Increasing Diversity in Open-Ended LLM Generation *Ongoing* [Writeup] [code]

Built a Hivemind Ladder pipeline to study why LLMs produce semantically similar answers to open-ended prompts and how to increase meaningful response diversity without sacrificing quality. Tested seven intervention levels, including temperature variation, cultural LoRA adapters, cultural prompt framing, few-shot cultural exemplars, cross-response n-gram repulsion, and DPP-based subset selection. Found that adapter-only and sampling-only methods produced little diversity gain, while culturally grounded few-shot prompting and generation-selection substantially improved semantic diversity. The full pipeline increased semantic diversity from 0.109 to 0.520 and Vendi score from 1.91 to 9.61 on Mistral-7B-v0.2.

Universal Lies: A Multi-Method Analysis of Hallucinations *with Rachit Das* [paper] [code]

Studied whether hallucinations in LLMs leave detectable internal signatures across five models from the Qwen, Llama, and Mistral families on TruthfulQA. Implemented attention analysis, hidden-state geometry, diagnostic probes, and targeted neuron ablations. Found that hallucinated generations attend less to question-region tokens in late-layer heads, while probes on hidden states reached peak AUC-ROC of 0.85 for hallucination detection. Ablating a sparse set of hallucination-associated H neurons substantially reduced the truthful/hallucinated geometric separation, suggesting these neurons are structurally involved in the observed representation.

Reimplementation: Internal Coherence Maximization for Persona Elicitation [Original paper] [code]

Reimplemented Internal Coherence Maximization (ICM), an unsupervised elicitation method that searches for label assignments that are mutually predictable and internally coherent under a pretrained model. Applied ICM to persona elicitation, where inferred labels can be used as a structured representation of a persona's values, beliefs, or preferences for guiding model behavior. Evaluated whether coherence-based elicitation recovers the target persona's values or instead reinforces pretrained model priors on culturally specific questions.

Engineering Projects

Blockchain-Based Dataset Registry [code]

Crystal, Blockchain, Cryptographic Hashing

Built a tamper-resistant dataset registry that generates cryptographic hashes from dataset metadata to detect duplicate or modified datasets before registration. Designed the system to improve dataset provenance, integrity checking, and reproducibility in collaborative ML workflows where multiple users may submit or update datasets.

CodeRoyale, Battle Royale Coding Platform
Node.js, Express, MongoDB, React, Elo Rating

[code]

Developed a competitive programming platform supporting real-time one-on-one coding challenges, user authentication, problem management, and a global leaderboard. Implemented an Elo-based rating system to dynamically update user rankings after matches and designed the backend APIs for challenge creation, result tracking, and leaderboard updates.

Credit App, Loan and Credit Management

[code]

Django, Django REST Framework, PostgreSQL, REST APIs

Built a full-stack credit and loan management application with secure API endpoints for customer data, loan records, and credit score calculation. Implemented backend logic for credit eligibility checks, database models for financial records, and RESTful endpoints for managing user and loan information.

Honors and Awards

Google Code Jam 2022, Round 1, top 20%

2022

National Stage Qualifier, International Mathematics Olympiad (IMO)

2019

3-star coder on CodeChef (max rating: 1650)

Technical Skills

Languages: Python, C, C++, JavaScript, SQL, Ruby, Crystal

ML / DL: PyTorch, Hugging Face Transformers, TransformerLens, PEFT/LoRA, scikit-learn, NumPy, pandas, Matplotlib

LLM Systems: LLM inference, fine-tuning workflows, hidden-state extraction, attention analysis, probing classifiers, evaluation pipelines

AI Safety / Interpretability: hallucination detection, mechanistic interpretability, representation geometry, value elicitation, pluralistic alignment, response diversity evaluation

Backend / Web: Django, Django REST Framework, Node.js, Express, React, REST APIs, MongoDB, PostgreSQL

Tools: Git, Docker, Google Colab, Kaggle, Linux, Jupyter, Overleaf